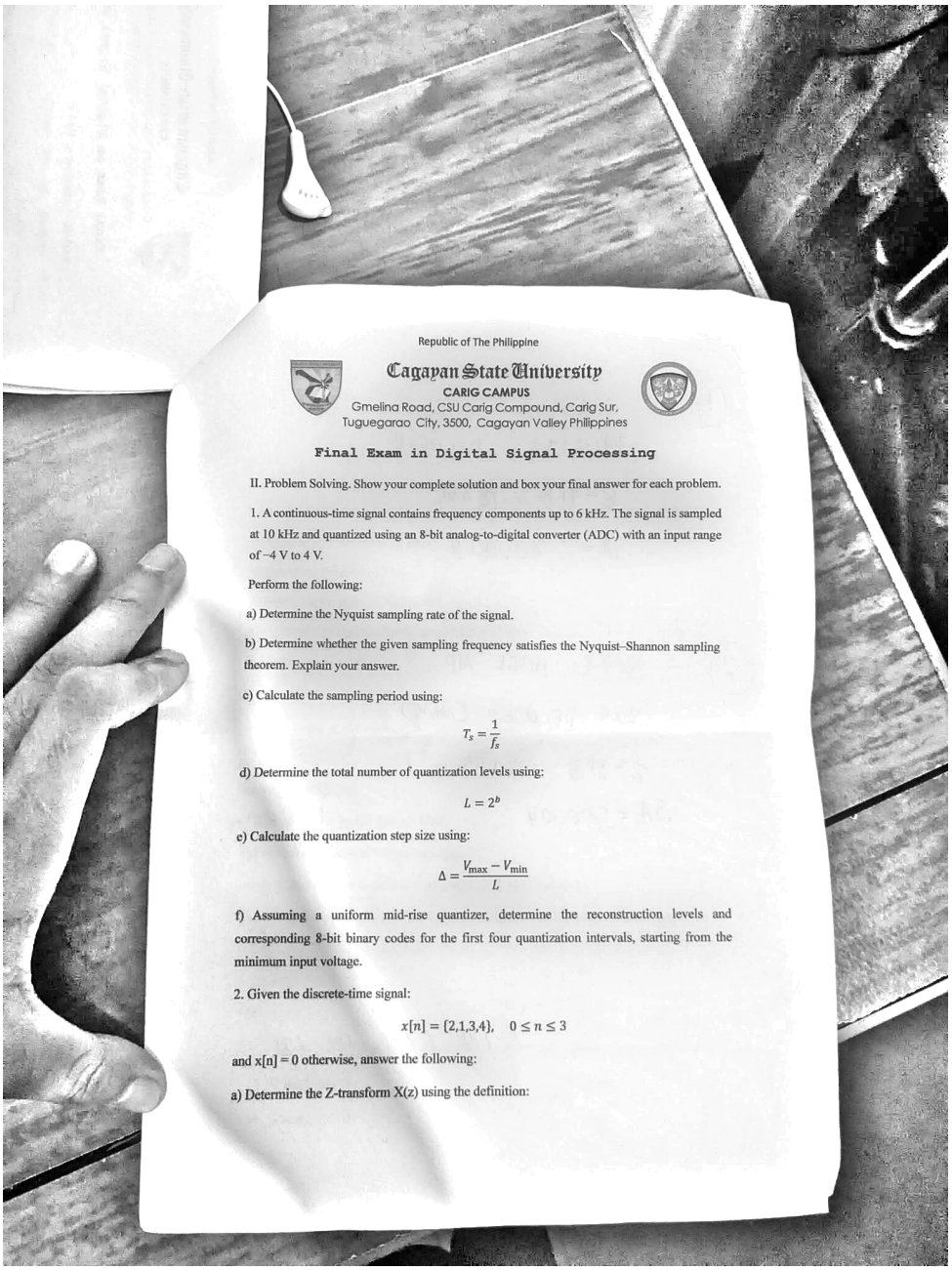
SmartBoard Session Report

Course: cpe14 - DSP
Session: Quantization
Date: 2026-10-24
Status: draft



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Final Exam in Digital Signal Processing

II. Problem Solving. Show your complete solution and box your final answer for each problem.

1. A continuous-time signal contains frequency components up to 6 kHz. The signal is sampled at 10 kHz and quantized using an 8-bit analog-to-digital converter (ADC) with an input range of −4 V to 4 V.

Perform the following:

a) Determine the Nyquist sampling rate of the signal.

b) Determine whether the given sampling frequency satisfies the Nyquist–Shannon sampling theorem. Explain your answer.

c) Calculate the sampling period using:

$$T_s = \frac{1}{f_s}$$

d) Determine the total number of quantization levels using:

$$L = 2^b$$

e) Calculate the quantization step size using:

$$\Delta = \frac{V_{max} - V_{min}}{L}$$

f) Assuming a uniform mid-rise quantizer, determine the reconstruction levels and corresponding 8-bit binary codes for the first four quantization intervals, starting from the minimum input voltage.

2. Given the discrete-time signal:

$$x[n] = \{2,1,3,4\}, \quad 0 \le n \le 3$$

and x[n] = 0 otherwise, answer the following:

a) Determine the Z-transform X(z) using the definition:

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of -4 V to4 V.